

Water, Acids, and Bases

Water is a common chemical substance on Earth.

Almost 75% of the planet is covered in water.

- 97% of that water is salt-water ocean

- 3% is freshwater

- 68% of all freshwater is locked in ice caps and glaciers (although that is changing)

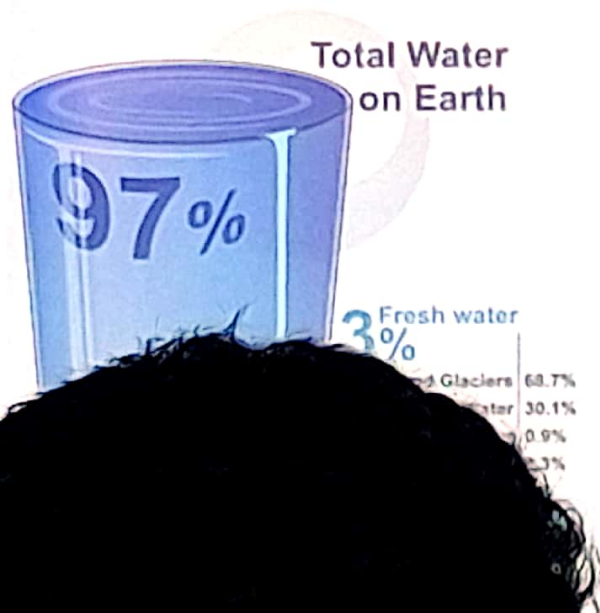
- 30% is ground water

- 2% is surface water

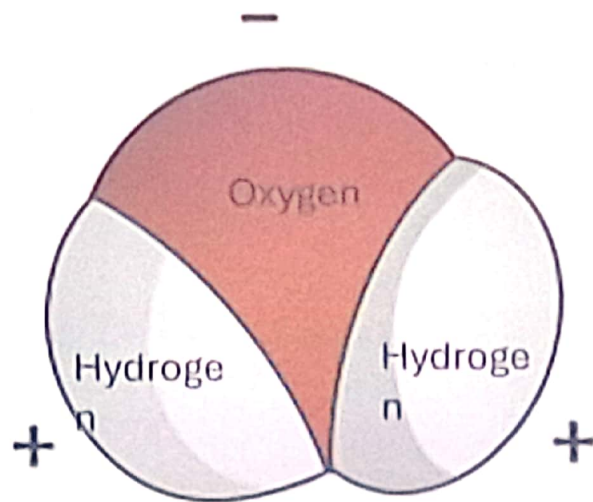
- 87% of the surface water is found in lakes

- 11% is located in swamps

- 2% is found in our rivers



Chemical Structure of Water

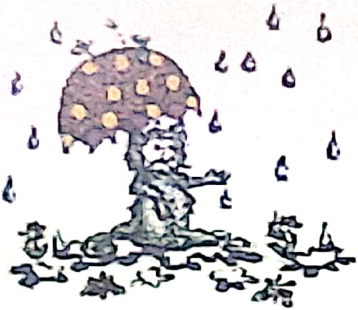


Water is made up of one atom of oxygen and two atoms of hydrogen.

The oxygen atom in a water molecule attracts electrons more strongly than the hydrogen atoms.

The result is that the oxygen side of the molecule is slightly negative while the hydrogen side is more positive.

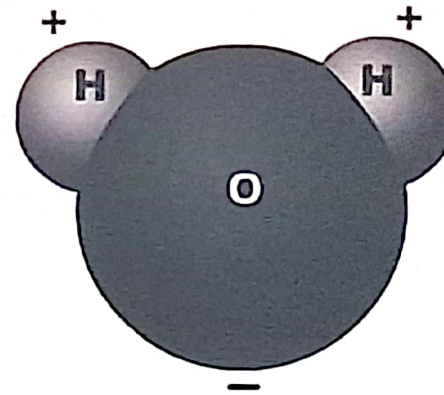
The difference in electrical charges on different sides of a molecule is called polarity. **Water is a polar molecule.**



W-A-T-E-R

More than 70% of the earth's surface is covered with water (H_2O)

- Atomic Structure = 2 hydrogen atoms bonded to 1 oxygen atom
- The hydrogen side: slightly positive (+)
- The oxygen side: slightly negative (-)



WATER: A Polar Molecule.....WHY?

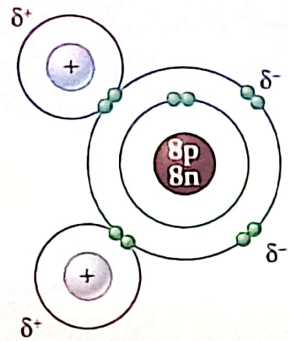
...Because there is an uneven distribution of electrons between the oxygen and hydrogen atoms

→ Results in water molecules being attracted to each other, forming bonds
Hydrogen bonds

Water Chemistry

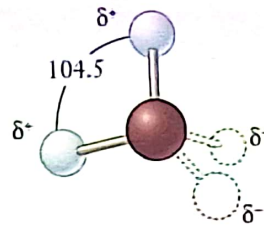
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Bohr Model



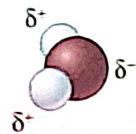
a.

Ball-and-Stick Model



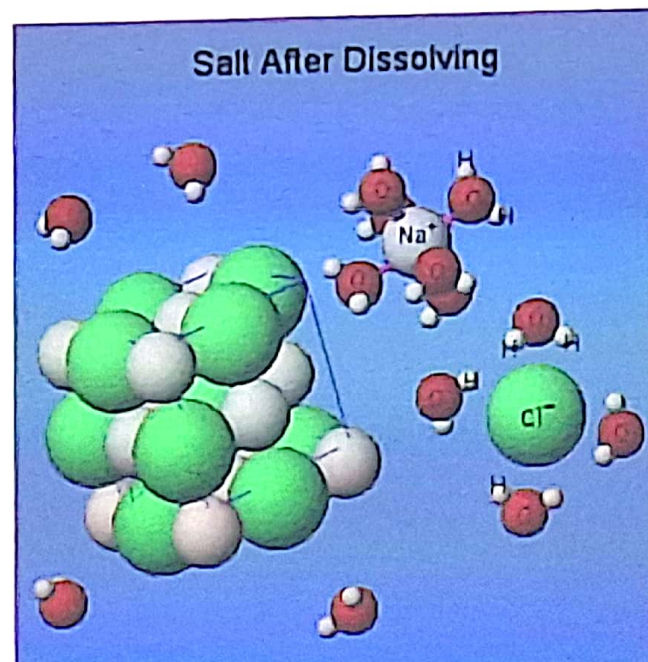
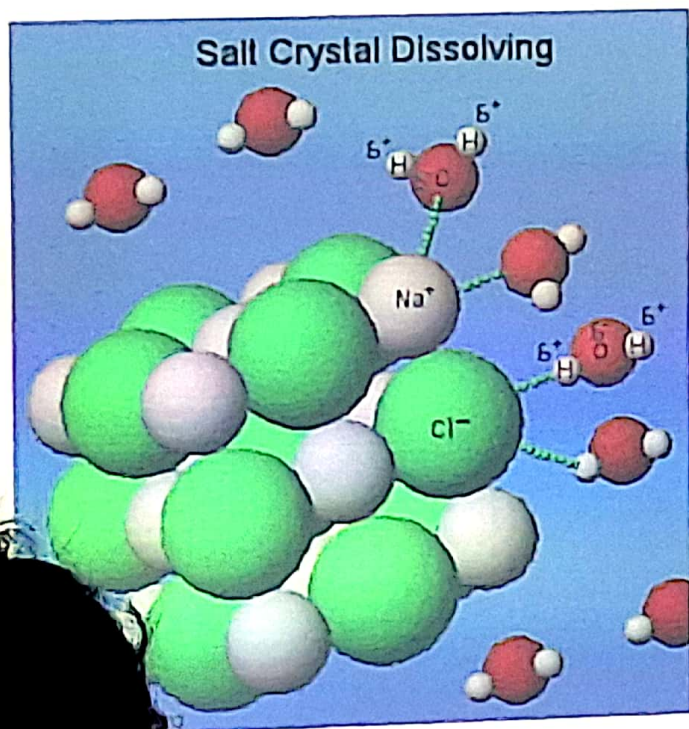
b.

Space-Filling Model



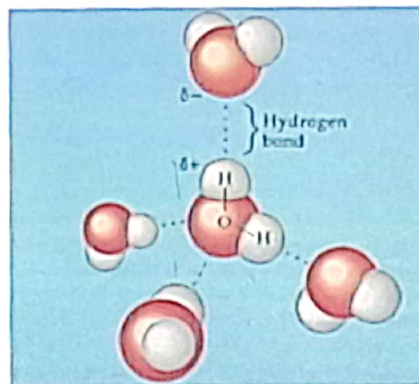
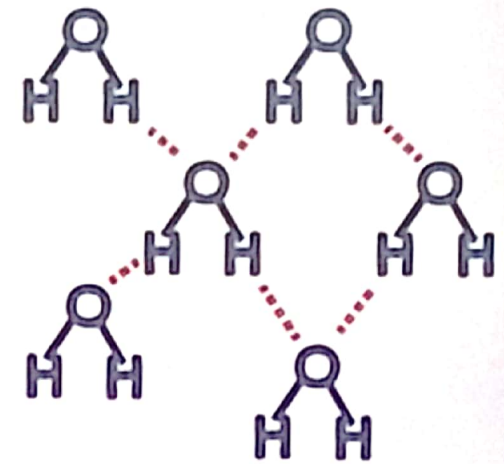
Why is Polarity So Important?

allows water to dissolve many substances including other polar molecules and ionic compounds



Water Molecules are Held Together by Hydrogen Bonds

- Hydrogen bonds are weak bonds between a hydrogen atom in one molecule and a negatively charged region of another molecule
- Hydrogen bonds are strong enough to cause water to cling to itself and other substances



W-A-T-E-R

All living organisms are dependent on water. The structure of water is the basis for its unique properties.

Properties of Water

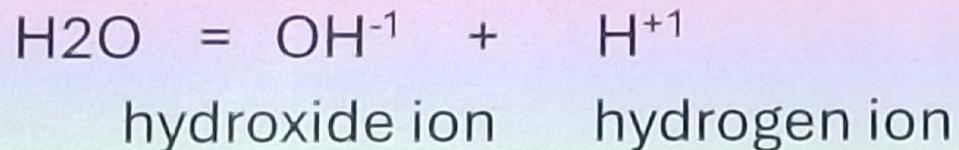
1. Water has a high specific heat.
 - A large amount of energy is required to change the temperature of water.
2. Water has a high heat of vaporization.
 - The evaporation of water from a surface causes cooling of that surface.
3. Solid water is less dense than liquid water.
 - Bodies of water freeze from the top down.
4. Water is a good solvent.
 - Water dissolves polar molecules and ions.

Properties of Water

1. Water organizes nonpolar molecules.

- hydrophilic: “water-loving”
- hydrophobic: “water-fearing”
- Water causes hydrophobic molecules to aggregate or assume specific shapes.

6. Water can form ions.



PROPERTIES OF:

Adhesion, specific heat, universal solvent, capillary action, surface

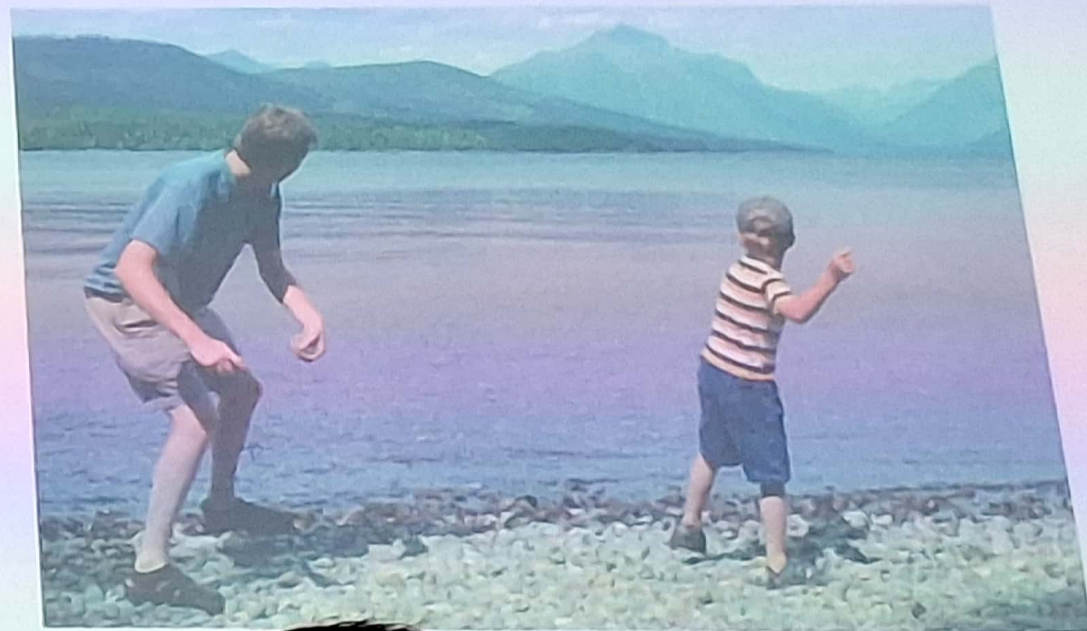
BECAUSE WATER IS A POLAR SUBSTANCE!

Many properties of water result from Hydrogen Bonding

Cohesion- attraction of molecules of the same substance (ex. water attracted to water)



Surface tension-
resistance of water
molecules at the surface
to a body of water ex.
bug "walking on water"

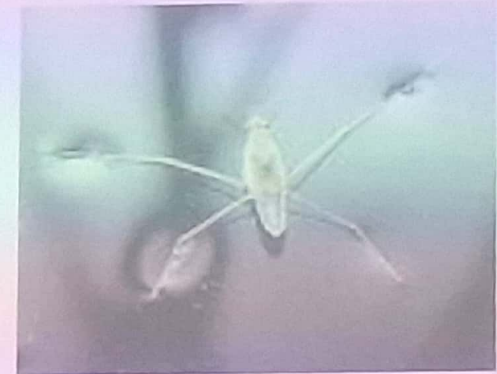


The tightness and resistance of water that is caused by
molecules near the surface makes the surface act like

Many properties of water result from Hydrogen Bonding

Cohesion- attraction of molecules of the same substance (ex. water attracted to water)

> Surface tension- cohesion of water molecules at the surface of a body of water ex. bug "walking on water"

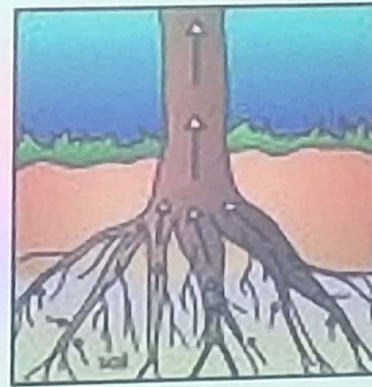


The force that holds the surface of water together is called surface tension. This force is caused by the attraction between water molecules. One another makes the

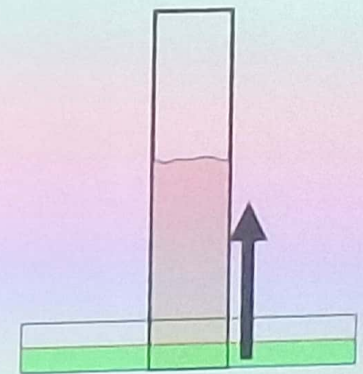
Many properties of water result from Hydrogen Bonding

Adhesion –attraction between molecules of different substances (ex. Water molecules attracted to a glass)

➤ Capillary Action- the adhesion of water molecules & another substance moving up against gravity



Cohesion + Adhesion



The combined force of attraction among water molecules and with the molecules of the surrounding material

Is Hydrogen Bonding related to Temperature?

- Hydrogen bonding makes water able to absorb a lot of heat before it gets hot and changes temperature because hydrogen bonds must break for the temperature to change
- This is called having a high specific heat.

Water also has a unique density

- Solid water is LESS DENSE than liquid water
ex. Ice floats (icebergs)
Ponds freeze from top down



WATER IS A UNIVERSAL SOLVENT

Water dissolves LOTS of things.

Remember:

SOLVENT = the thing doing the dissolving

SOLUTE = the thing that dissolves away

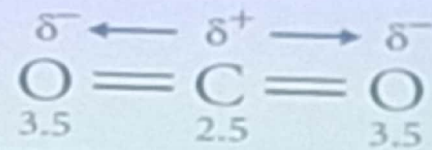
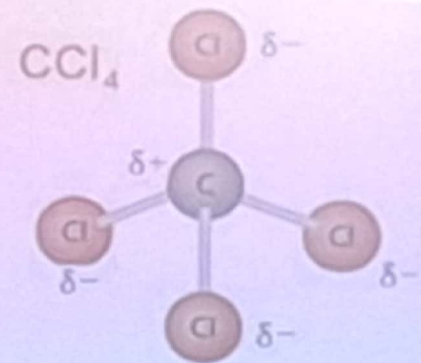


Other polar molecules can be dissolved by water.

Non-polar substances DO NOT dissolve in water.

Can you think of some examples of non-polar substances?

Non-polar substances have evenly distributed electrons, examples include hydrocarbons (oil, gasoline, methane, alkanes like pentane), diatomic gases (oxygen (O_2), nitrogen (N_2), noble gases (helium, argon), and symmetrical molecules like carbon dioxide (CO_2) and carbon tetrachloride (CCl_4). These substances don't dissolve well in water because they lack positive/negative poles, unlike polar substances.



Density of Water

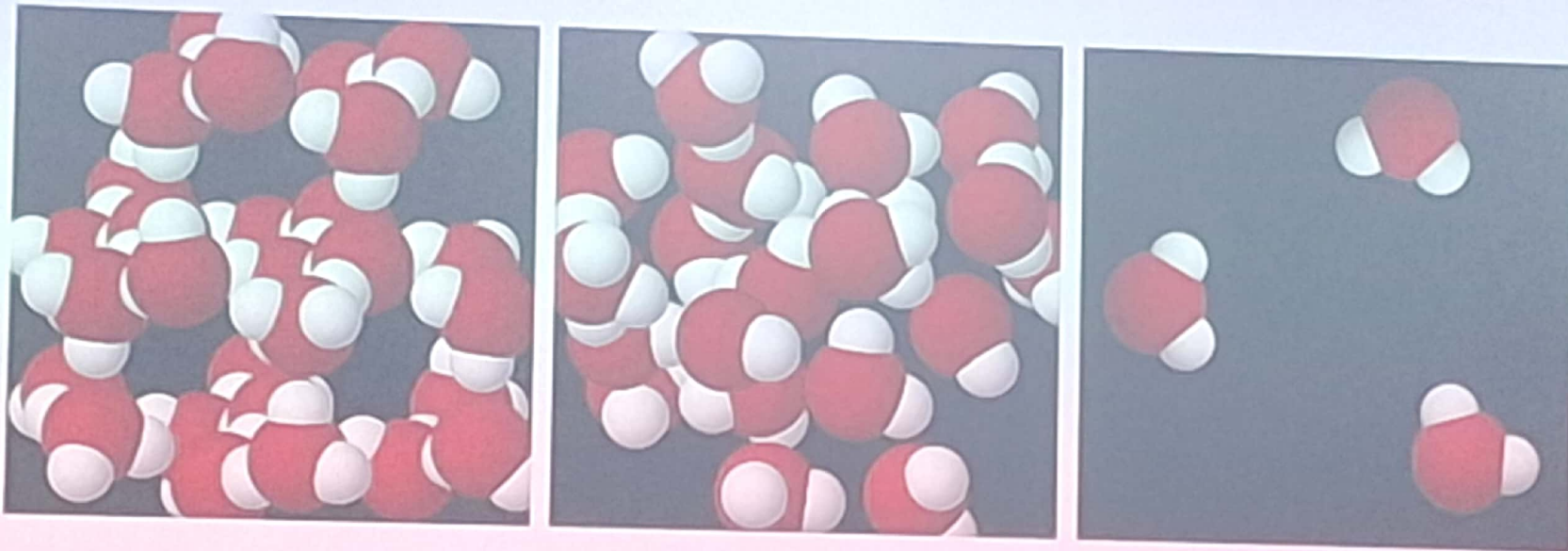
- Most dense at 4°C
- Contracts until 4°C
- Expands from 4°C to 0°C



The density of water:

1. Prevents water from freezing from the bottom up.
2. Ice forms on the surface first—the freezing of the water releases heat to the water below creating insulation.
3. Makes transition between season less abrupt.

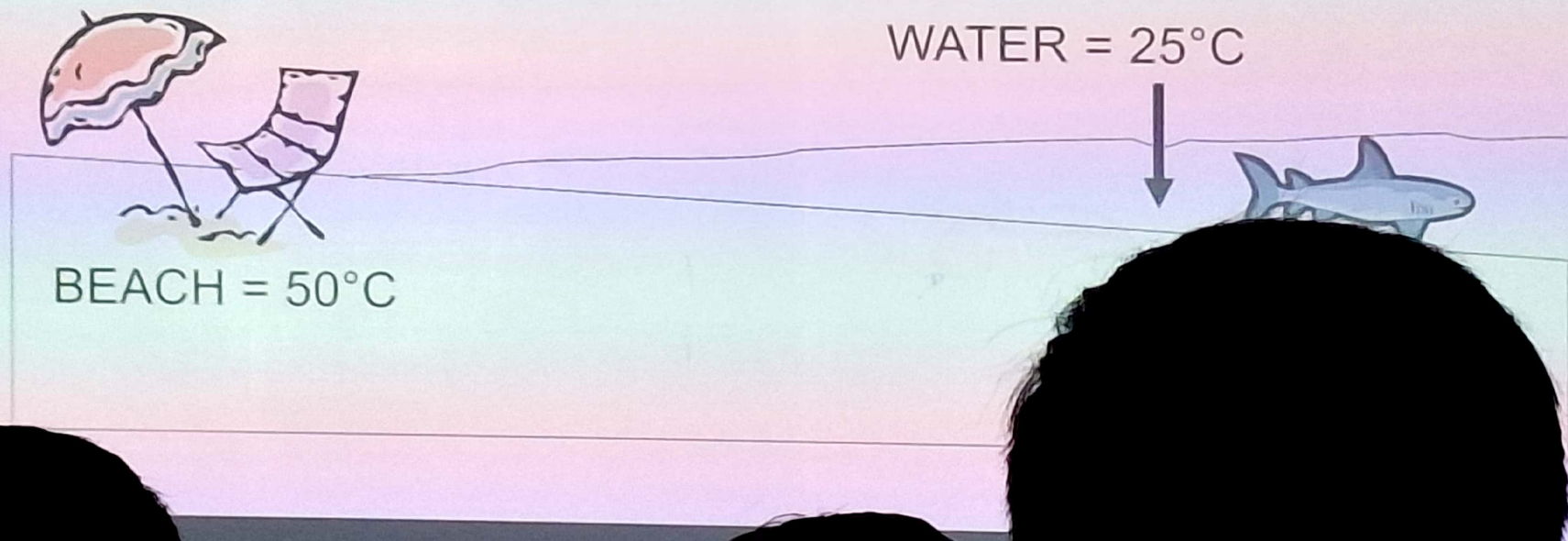
Ice, water, vapor



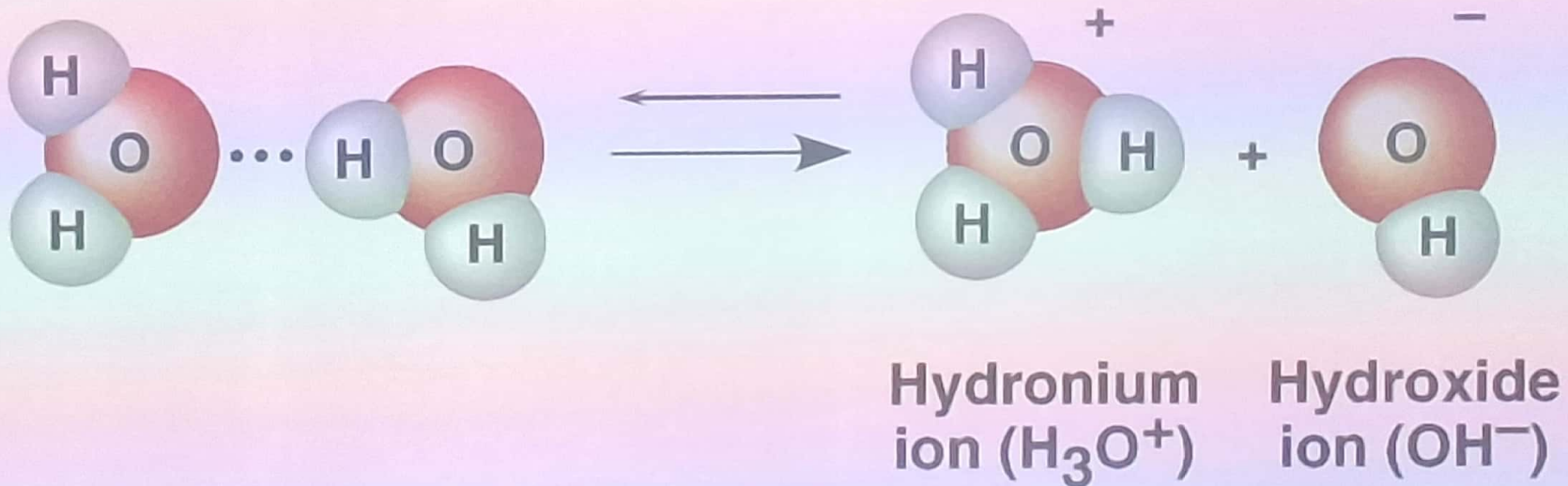
- When water reaches 0°C , water becomes locked into a crystalline lattice with each molecule bonded to to the maximum of four partners.
- As ice starts to melt, some of the hydrogen bonds break and some water molecules can slip closer together than they can while in the ice state.
- Ice is about 10% less dense than water at 4°C .

SPECIFIC HEAT

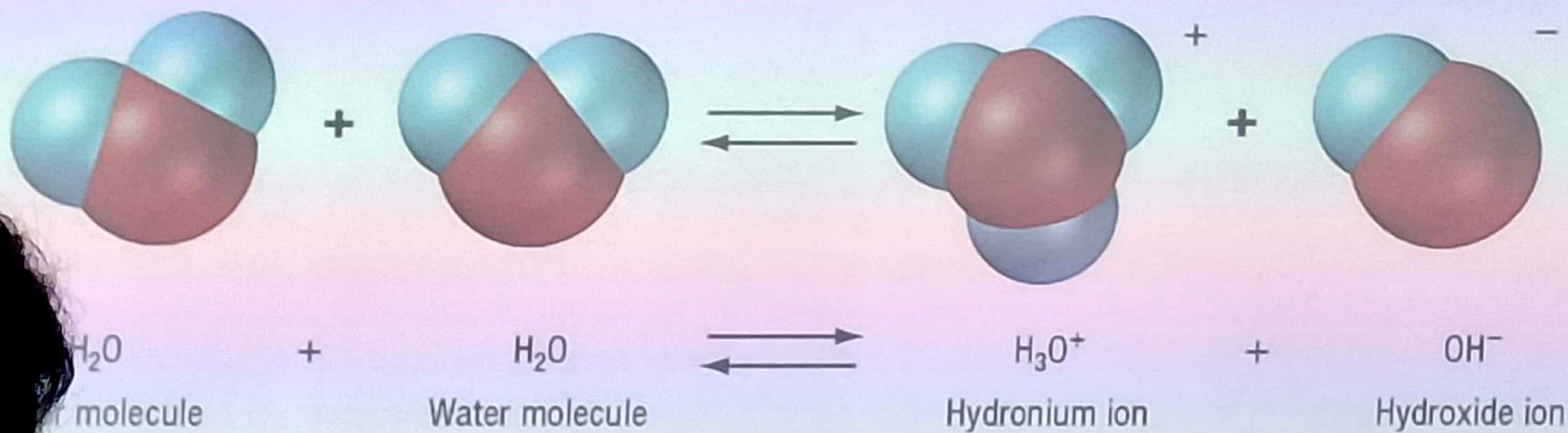
- Specific Heat = the amount of energy needed to increase the temperature of something 1 degree C.
- Water has a really HIGH specific heat
- That means it takes a lot of energy for water to increase its temperature.
- This is because of the **STRONG ATTRACTION** between water molecules.



“Dissociation” of water



Acid, Base and pH



Buffers

- **Buffers** resist changes to the pH of a solution when H^+ or OH^- is added to the solution.
- Buffers accept hydrogen ions from the solution when they are in excess and donate hydrogen ions when they have been depleted.

